Picking Winners or Building Them?

Central and Local Government Venture Capital in Chinese Semiconductors

Ming-yen Ho
UC Berkeley, Haas School of Business

New Thinking in Industrial Policy, Columbia University, November 6, 2026

Industrial Policy Is Back. How Should It Be Run?

- Chips, electric vehicles, shipping: governments are again backing strategic sectors (Juhász et al., 2024, Goldberg et al., 2024)
- The debate is mostly about **whether** to intervene and **which sectors**. We know much less about **how**
- The track record: support often went to weak firms that never improved (Krueger, 1990, Beason and Weinstein, 1996, Lerner, 2009)
- **This paper**: within a sector the state has already committed to, **which level of government should choose the firms?**

China Invested in a Champion. Who Built It?

- 2016: the **local** Hefei city government's fund invests in memory-chip maker CXMT
- 2020: the **central** government's "Big Fund" and a **private** VC follow
- 2026: CXMT's IPO makes it the most valuable listed company in mainland China

Was the venture capitalist Beijing, or Hefei?


China | Power law

Xi Jinping, pioneering venture capitalist

CXMT's monster initial public offering is a money-maker for the state. Its success may not be repeated

ILLUSTRATION: THE ECONOMIST / GETTY IMAGES

Source: *The Economist*, 27 July 2026.

China Invested in a Champion. Who Built It?

- 2016: the **local** Hefei city government's fund invests in memory-chip maker CXMT
- 2020: the **central** government's "Big Fund" and a **private** VC follow
- 2026: CXMT's IPO makes it the most valuable listed company in mainland China

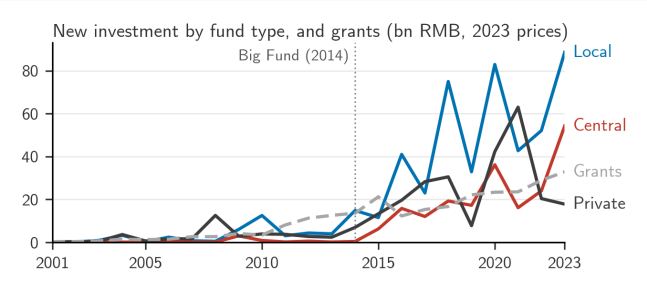
Was the venture capitalist Beijing, or Hefei?



Source: *The Economist*, 27 July 2026.

Local Governments Are China's Biggest Chip Investors

- Since 2014, Chinese governments fund chips mainly through **venture capital (VC) funds**: they take equity stakes, and professional fund managers pick the firms
- **Central**: the national "Big Fund" (2014). **Local**: provincial and city funds
- **Local** funds invested about $66B in 2014–23, the majority of China's chip industrial-policy spending; for scale, the US CHIPS Act promises $52.7B



Who Should Allocate? A Classic Trade-off

Central government

+ National view: can back the best firms anywhere in the country
– Far from most firms, with many firms to oversee

Local governments

– Home bias: officials are judged on local growth, so they back the best firm *at home*, even if it is weak nationally (Li and Zhou, 2005, Xu, 2011)
+ Close to firms: local knowledge, and can be hands-on

Does local governments' worse picking get offset by better building?

Decentralization as delegation to informed but self-interested local governments (Oates, 1999)

Who Should Allocate? A Classic Trade-off

Central government

+ National view: can back the best firms anywhere in the country
– Far from most firms, with many firms to oversee

Local governments

– Home bias: officials are judged on local growth, so they back the best firm *at home*, even if it is weak nationally (Li and Zhou, 2005, Xu, 2011)
+ Close to firms: local knowledge, and can be hands-on

Does local governments' worse picking get offset by better building?

Decentralization as delegation to informed but self-interested local governments (Oates, 1999)

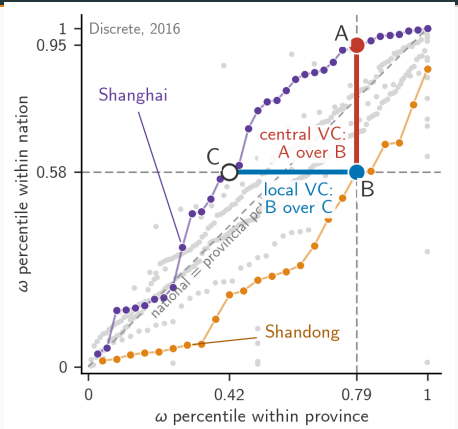
What I Find

1. **Picking**: **central** and **private** VCs back the nation's most productive firms; **local** funds back smaller, less productive home champions
2. **Building**: **local**-backed firms become about **34% more productive** after investment; no detectable change for **central**- or **private**-backed firms
3. **How**: **local** funds are nearby, hold fewer firms, stay longer, and reshape boards. Board and management changes account for **44%** of the **local**–**central** gap
4. **Whole industry**: in a model with import competition, 2023 domestic chip sales would be **5.5% lower** without government VC, and **2.4% lower** if the **central** fund made the **local** funds' investments

Data: Following the Money and the People

- **Business registry** for the universe of Chinese firms: every shareholder and stake over time
  - trace each fund up its ownership chain to a central, provincial, or city government
  - 2014–23: **local** funds invested in 1,645 firms, the **central** fund in 546; **private** VCs made 1,397 investments
- **Financials** of 6,067 semiconductor firms, 1998–2024
- **Boards and management**: who sat on each board and when, with biographies (education, careers) coded by LLMs
- **Innovation and favors**: patents, chip-design registrations, plant-level process technology, procurement contracts, subsidies

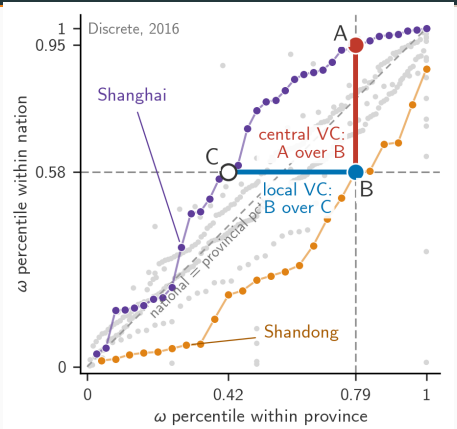
Central Picks National Champions; Local Picks Home Champions



Discrete-device makers, 2016. Horizontal: productivity rank at home; vertical: rank in the nation. Productivity (ω): sales from given capital and workers.

- Both tiers set **industry-sales targets**, so each backs the most productive firms *in its own jurisdiction*
- Same home rank (79th percentile): Shanghai's A is 95th nationally, Shandong's B only 58th. The **central** fund prefers **A**
- Same national rank (58th): Shandong's B is a home champion; Shanghai's C is not. Shandong's **local** fund prefers **B**
- Regressions with rich firm controls: national rank predicts **central** and **private** investment; home rank predicts **local** investment Backup

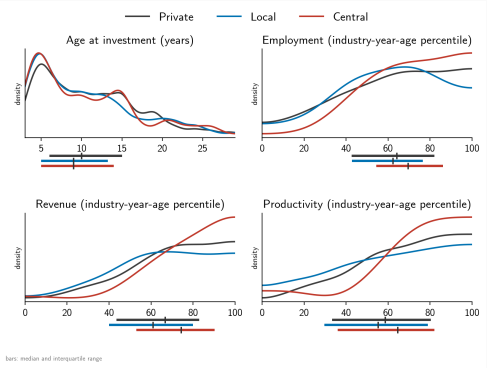
Central Picks National Champions; Local Picks Home Champions



Discrete-device makers, 2016. Horizontal: productivity rank at home; vertical: rank in the nation. Productivity (ω): sales from given capital and workers.

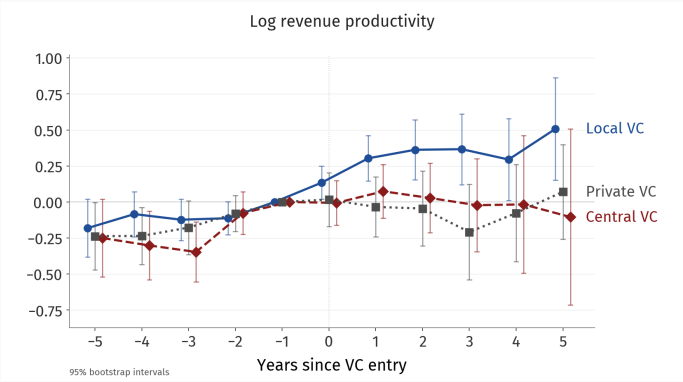
- Both tiers set **industry-sales targets**, so each backs the most productive firms *in its own jurisdiction*
- Same home rank (79th percentile): Shanghai's A is 95th nationally, Shandong's B only 58th. The **central** fund prefers **A**
- Same national rank (58th): Shandong's B is a home champion; Shanghai's C is not. Shandong's **local** fund prefers **B**
- Regressions with rich firm controls: national rank predicts **central** and **private** investment; home rank predicts **local** investment Backup

So Local Funds Back Smaller, Less Productive Firms



- Firms in the year before their first VC investment, ranked against same-sector firms of the same age. **Central** picks are the largest and most productive
- From Beijing's view this looks like **misallocation**, unless **local** funds make their firms better

Local-Backed Firms Become More Productive; Others Do Not



Local: about **+34%** on average from the investment year through year 5 (0.3 on the vertical axis ≈ +35%). **Central**-backed firms were already improving *before* investment: picked on recent success

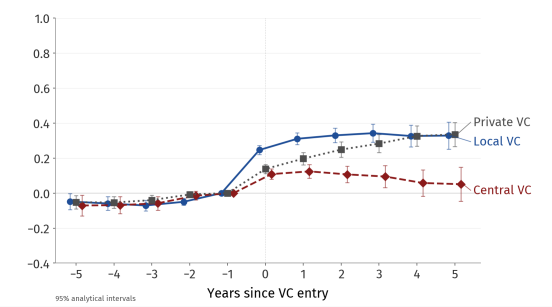
Backup

The Gains Are Real Output, Not Just Favors

- **Sales**: revenue is about 60% higher three years after **local** investment
- **Innovation**: more invention patents and registered chip designs after **local** investment than after **central** or **private** investment
- **Not protectionism**: **local**-backed firms do not win more government procurement contracts. If anything, **central**-backed firms do
- **Capital**: all three investor types expand their firms' capital (three years on: +38% **local**, +45% **central**)
- **First-time investments only**: no differences before investment, and the same **local** advantage afterward Backup

Local Funds Reshape Boards and Management

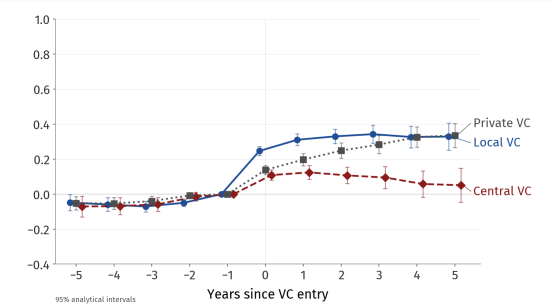
Number of people on the board or management team (log points vs. comparison firms)



- After **local** investment, boards and management teams grow about **30%**, right away
- New members more often hold graduate degrees, foreign degrees, and finance experience Backup
- **Private**: similar growth, but slower. **Central**: a smaller rise that fades
- Example: Shenzhen Capital Group (a city fund) helped laser-chip startup Xinchen design its board, set up performance pay and budgets, and hire finance staff

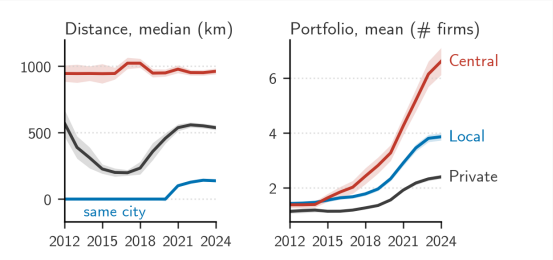
Local Funds Reshape Boards and Management

Number of people on the board or management team (log points vs. comparison firms)



- After **local** investment, boards and management teams grow about **30%**, right away
- New members more often hold graduate degrees, foreign degrees, and finance experience Backup
- **Private**: similar growth, but slower. **Central**: a smaller rise that fades
- Example: Shenzhen Capital Group (a city fund) helped laser-chip startup Xinchen design its board, set up performance pay and budgets, and hire finance staff

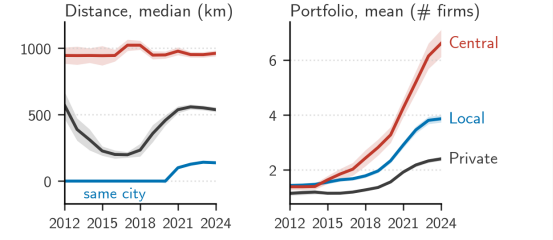
Local Funds Are Close, Focused, and Patient



Shaded: ±1 SE (bootstrap SE for the median)

- **Close**: the median **local**-backed firm is in the fund's own city (within about 150 km since 2021); the **central** fund is about 1,000 km away
- **Focused**: **local** funds oversee fewer firms each
- **Hands-on**: **local**-backed firms have 0.51 fund directors on their boards vs. 0.29 (**central**). Within a firm, more distance and bigger portfolios mean fewer
- **Patient**: 37% of **private** VCs exit within five years, vs. 23% (**local**) and 22% (**central**) Backup

Local Funds Are Close, Focused, and Patient



Shaded: ±1 SE (bootstrap SE for the median)

- **Close**: the median **local**-backed firm is in the fund's own city (within about 150 km since 2021); the **central** fund is about 1,000 km away
- **Focused**: **local** funds oversee fewer firms each
- **Hands-on**: **local**-backed firms have 0.51 fund directors on their boards vs. 0.29 (**central**). Within a firm, more distance and bigger portfolios mean fewer
- **Patient**: 37% of **private** VCs exit within five years, vs. 23% (**local**) and 22% (**central**) Backup

How Much Does Hands-On Management Explain?

Channel	Share of gap
Board and management	**44%**
Non-equity subsidies	10%
Procurement	8%
Patents	3%
Shareholder capital	−3%
All measured channels	62%
Unexplained	38%

Gap between **local** and **central** productivity effects, 3–5 years after investment.

- Board and management changes are the **largest** measured channel; money and favors explain little
- The unexplained part may be **local information**:
  - **Local** funds are more likely to invest in firms linked to their managers through alumni and career ties
  - **Central** and **private** investors tend to follow **local** funds into firms

How Much Does Hands-On Management Explain?

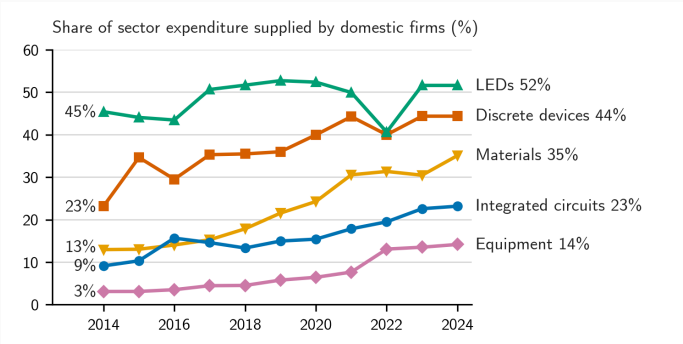
Channel	Share of gap
Board and management	**44%**
Non-equity subsidies	10%
Procurement	8%
Patents	3%
Shareholder capital	−3%
All measured channels	62%
Unexplained	38%

Gap between **local** and **central** productivity effects, 3–5 years after investment.

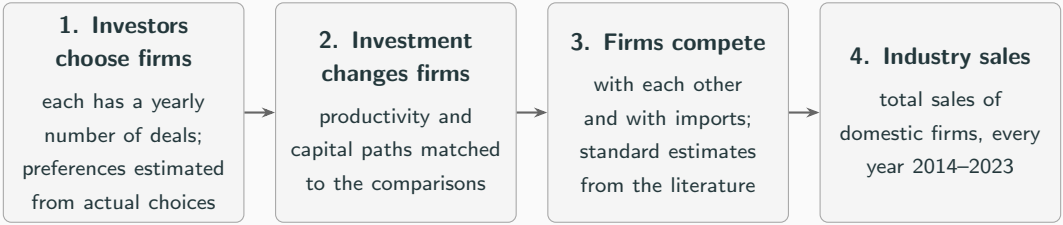
- Board and management changes are the **largest** measured channel; money and favors explain little
- The unexplained part may be **local information**:
  - **Local** funds are more likely to invest in firms linked to their managers through alumni and career ties
  - **Central** and **private** investors tend to follow **local** funds into firms

Gains Could Come From Domestic Rivals, or From Imports

- One firm's gain may just be a domestic rival's loss. Then the industry gains nothing
- But imports meet most of China's chip demand: domestic firms supply 23% of integrated circuits and 14% of equipment (2024), so gains can come at the expense of imports (Rotemberg, 2019)



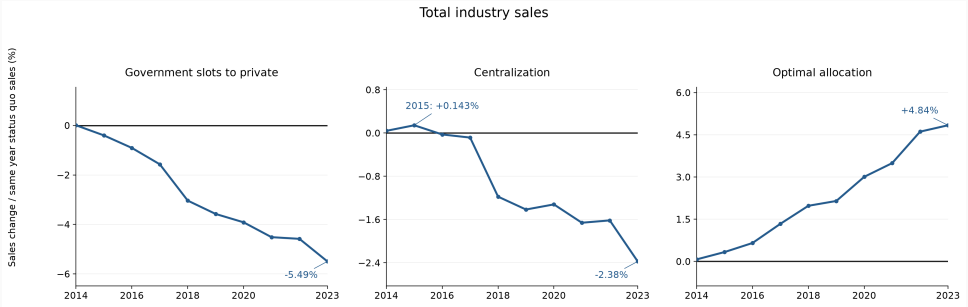
A Model to Add It Up



Counterfactuals: same total number of deals each year; each investor keeps its preferences and its effect on firms. Only *who gets the deals* changes:

- **No government VC**: every government deal goes to **private** VC
- **Centralization**: every **local** deal goes to the **central** fund
- **Best allocation**: a planner splits government deals between **central** and **local** funds to maximize 2023 sales Backup

Decentralization Raised Industry Sales



- **No government VC**: 2023 domestic sales **5.5% lower**
- **Centralization**: slightly ahead at first, behind from 2016 on; **2.4% lower** by 2023
- **Best allocation** shifts deals *toward* **local** funds: **4.8% higher**
- Why: centralization's better picks add only 0.4 points; losing **local** funds' productivity gains costs 6.2 points Backup

Takeaways

- **Who allocates matters**: the same sector and the same instrument gave different results depending on the level of government
- **Local** funds **picked weaker firms but built them**: faster growth outweighed the cost of backing ex ante weaker firms
- **Equity, not just grants**: ownership brought board seats, management talent, and patience
- Holds even if local gains reflect *knowing* which firms will grow rather than *helping* them grow Backup
- **Open question**: how to keep local knowledge and engagement while limiting favoritism and local protectionism

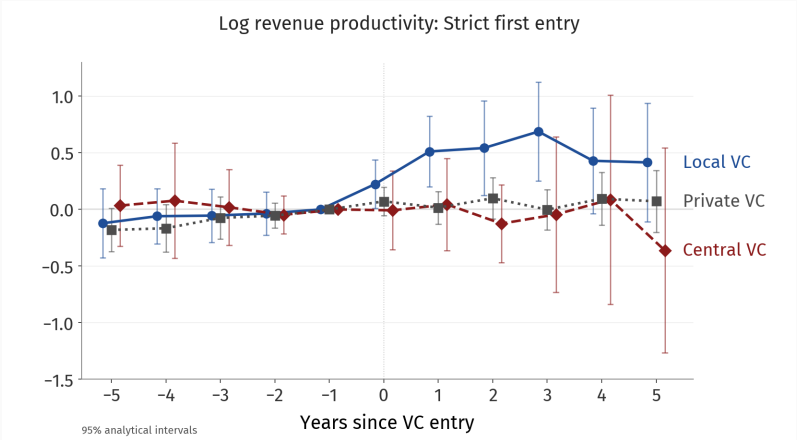
Thank you!

Paper: homervyn.github.io/files/Ho_JMP.pdf

mho@berkeley.edu

Productivity: First-Time VC Investments Only

Treated: firms receiving their *first* VC investment, from one investor type only. Comparison: same-sector firms with no VC yet. 510 **local**, 72 **central**-only, 933 **private** first investments.



How the Comparison Works

- Firms can receive money from all three investor types, at different times
- For each investor type and year: **treated** = firms it first invests in that year; **comparison** = same-sector firms it has not yet invested in, with the same history of other investors, and no new investor in between
- Average these before-and-after comparisons across years and histories (de Chaisemartin and D'Haultfœuille, 2023)
- Productivity: from production functions estimated separately for six semiconductor sectors (Ackerberg et al., 2015)
- Robust to two-way fixed effects, value-added productivity, and a translog production function (paper appendix)

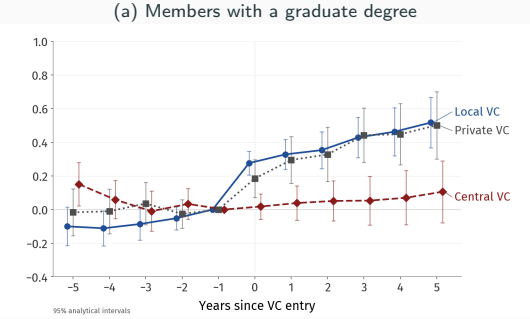
Who Invests in Whom? Regression Estimates

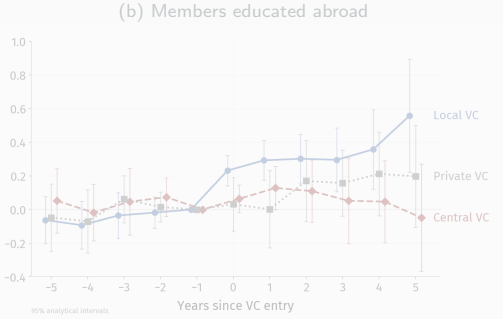
Outcome: 1(first investment by this investor type next year), among firms it has not yet invested in. Sector-by-year fixed effects; column (2) adds technology, size, ownership, board, subsidy and procurement controls.

	Central VC		Local VC		Private VC	
	(1)	(2)	(1)	(2)	(1)	(2)
National TFPR percentile (0–1)	**0.012****	**0.012****	-0.004	-0.005	**0.018*****	**0.018*****
	(0.006)	**(0.005)**	(0.006)	(0.006)	**(0.007)**	**(0.006)**
Provincial TFPR percentile (0–1)	-0.006	-0.008	**0.014****	**0.010***	0.006	0.002
	(0.005)	(0.005)	**(0.006)**	**(0.006)**	(0.006)	(0.006)
Mean outcome	0.01	0.01	0.02	0.02	0.01	0.01
N	22,533	22,472	23,964	23,902	20,874	20,874
R^2	0.04	0.04	0.04	0.05	0.02	0.04
Clusters (prov. × ind.)	175	175	177	177	174	174
Ind. × year FE	Y	Y	Y	Y	Y	Y
Firm and policy controls	N	Y	N	Y	N	Y

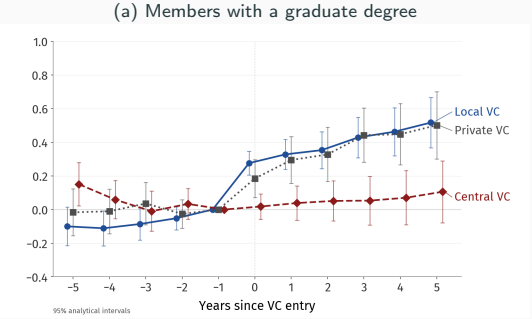
Bottom to top of the national ranking: +1.2 pp for **central** (base ≈1%). Bottom to top of the provincial ranking: +1.0 to 1.4 pp for **local** (base ≈2%).

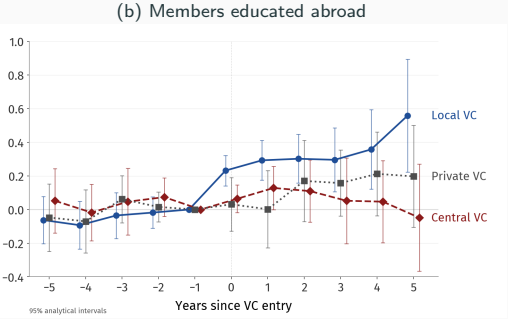
Who Joins the Board



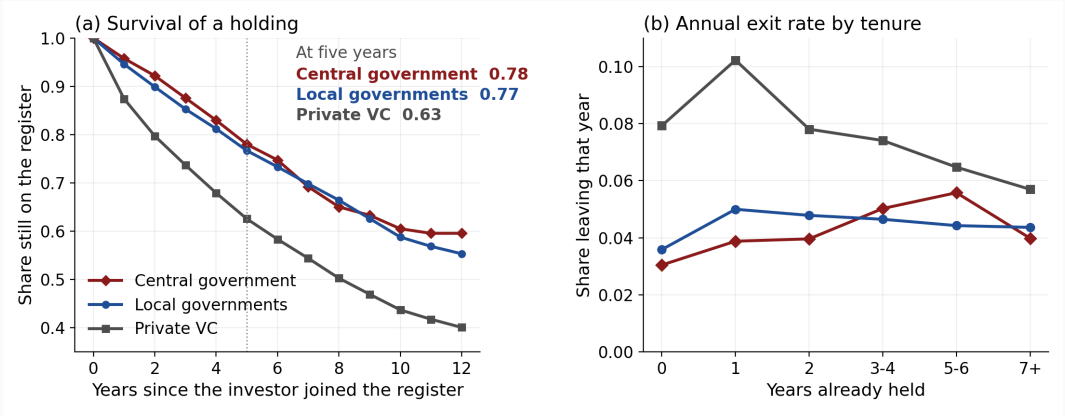


Who Joins the Board





Private VCs Exit Sooner

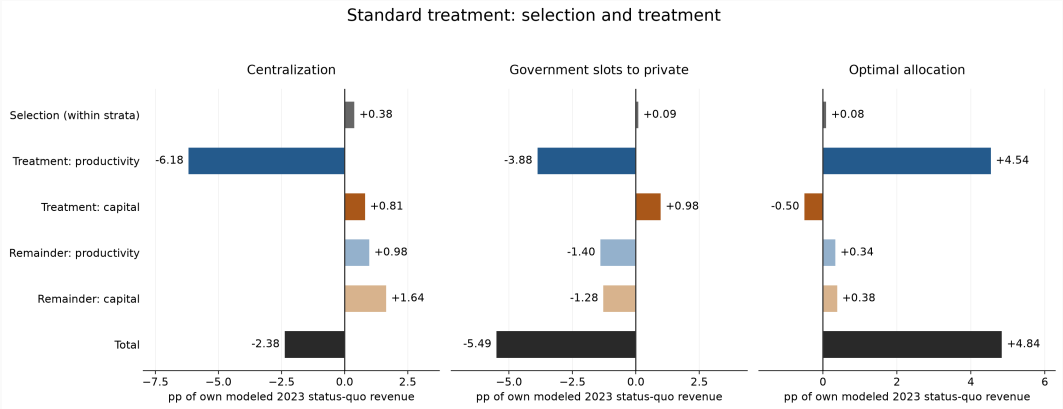


Private VCs exit more once the firm goes public. Being based in the firm's city keeps government funds invested longer, but not private VCs.

Model Ingredients (100 Simulated Histories, 2014–2023)

Ingredient	Value	Source
Investors' choices	estimated	observed first investments: firm characteristics, alumni ties, other investors
Effects on productivity and capital	estimated	matched to the event-study comparisons
Production	estimated by sector	production-function estimates
Domestic vs. import shares	data	firm sales and customs trade flows
Domestic chips vs. imports	substitution elasticity 3 (2–3.48)	Feenstra et al. (2018)
Among domestic chips	substitution elasticity 5 (4.57–5.48)	Amiti et al. (2020); Fan et al. (2020)
Held fixed	—	wages, import prices, downstream spending, firm entry and exit

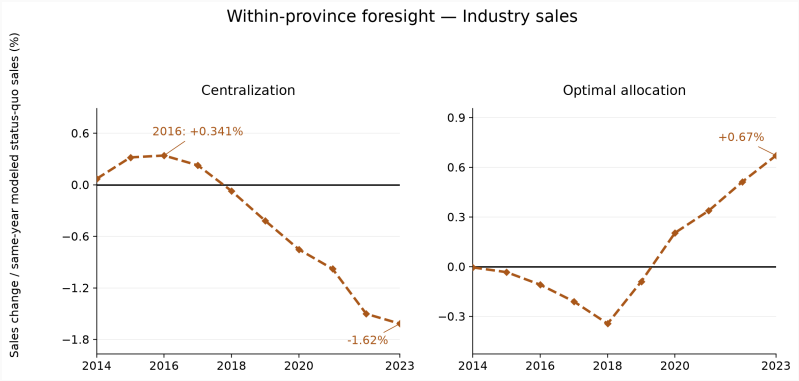
Picking vs. Building: Decomposing 2023 Sales



Selection: who receives the deals. Treatment: investors' different effects on productivity and capital. Remainder: how many firms are funded, in which sectors and years.

If Local Gains Were Pure Information

Alternative model: no investor changes productivity; **local** funds instead see a noisy signal of their firms' future productivity, calibrated to match the same comparisons.



Centralization is ahead through 2017, then falls behind: 1.6% lower sales by 2023.

References

Ackerberg, D. A., K. Caves, and G. Frazer (2015): "Identification Properties of Recent Production Function Estimators," *Econometrica*, 83, 2411–2451.

Amiti, M., M. Dai, R. C. Feenstra, and J. Romalis (2020): "How Did China's WTO Entry Affect U.S. Prices?" *Journal of International Economics*, 126, 103339.

Beason, R. and D. E. Weinstein (1996): "Growth, Economies of Scale, and Targeting in Japan (1955–1990)," *Review of Economics and Statistics*, 78, 286–295.

de Chaisemartin, C. and X. D'Haultfœuille (2023): "Two-Way Fixed Effects and Differences-in-Differences Estimators with Several Treatments," *Journal of Econometrics*, 236, 105480.

Fan, H., Y. A. Li, S. Xu, and S. R. Yeaple (2020): "Quality, Variable Markups, and Welfare: A Quantitative General Equilibrium Analysis of Export Prices," *Journal of International Economics*, 125, 103327.

Feenstra, R. C., P. Luck, M. Obstfeld, and K. N. Russ (2018): "In Search of the Armington Elasticity," *The Review of Economics and Statistics*, 100, 135–150.

Goldberg, P. K., R. Juhász, N. J. Lane, G. Lo Forte, and J. Thurk (2024): "Industrial Policy in the Global Semiconductor Sector," Working Paper 32651, National Bureau of Economic Research.

Juhász, R., N. Lane, and D. Rodrik (2024): "The New Economics of Industrial Policy," *Annual Review of Economics*, 16, 213–242.

Krueger, A. O. (1990): "Government Failures in Development," *Journal of Economic Perspectives*, 4, 9–23.

Lerner, J. (2009): *Boulevard of Broken Dreams: Why Public Efforts to Boost Entrepreneurship and Venture Capital Have Failed—and What to Do About It*, Princeton, NJ: Princeton University Press.

Li, H. and L.-A. Zhou (2005): "Political Turnover and Economic Performance: The Incentive Role of Personnel Control in China," *Journal of Public Economics*, 89, 1743–1762.

Oates, W. E. (1999): "An Essay on Fiscal Federalism," *Journal of Economic Literature*, 37, 1120–1149.

Rotemberg, M. (2019): "Equilibrium Effects of Firm Subsidies," *American Economic Review*, 109, 3475–3513.

Xu, C. (2011): "The Fundamental Institutions of China's Reforms and Development," *Journal of Economic Literature*, 49, 1076–1151.